Variational Quantum Circuits for Pesticide Environmental Fate Prediction: A Comparative Study with Classical Machine Learning on the EU SPIN Database

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Abstract

Predicting pesticide environmental fate properties specifically soil degradation half-life (DegT50) and organic carbon sorption coefficient (K_{oc})—is essential for regulatory risk assessment under EC 1107/2009. Classical QSAR models achieve moderate predictive power for K_{oc} but struggle with DegT50 due to its dependence on complex soil microbial processes. This work investigates whether variational quantum circuits (VQCs) can capture non-linear molecular descriptor relationships that elude classical regressors. We encode 17 physicochemical descriptors for 111 EU-registered pesticides into a 12-qubit IQP-style quantum circuit with data re-uploading and 8 variational layers (301 trainable parameters). Leave-one-out cross-validation benchmarks reveal that Random Forest (RF) achieves $R^2 = 0.194$ for DegT50 and 0.766 for K_{oc}, while Gradient Boosting (GB) reaches 0.223 and 0.779, respectively. The VQC underperforms classical baselines on DegT50 ($R^2 < 0$, 5-fold CV), consistent with severe overfitting expected from the unfavorable parameter-to-sample ratio ($N_{\text{params}}/N_{\text{data}} \approx 2.7$). A hybrid stacking framework blending QML and RF predictions is introduced but remains RF-dominated ($\alpha \approx 0$). We also assess adverse outcome pathway (AOP) features derived from AOP-Wiki data as additional descriptors, finding no improvement over structural features alone. These results provide a rigorous benchmark for near-term quantum machine learning applied to regulatory chemistry and highlight the critical role of dataset size in VQC generalization.

Keywords: quantum machine learning, variational quantum circuits, pesticide fate, DegT50, Koc, QSAR, regulatory chemistry, EU SPIN

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1 Introduction

Pesticide environmental fate assessment is a cornerstone of EU regulatory approval under Regulation (EC) No 1107/2009 (ec1, 2009). Two properties dominate the risk evaluation: the soil degradation half-life (DegT50), governing persistence, and the organic carbon sorption coefficient (K_{oc}), governing mobility and leaching potential. Together, these parameters feed into the FOCUS groundwater models (foc, 2006) that determine whether a substance can be approved for agricultural use.

Experimental determination of DegT50 follows OECD Guideline 307 (oec, 2002), requiring months-long incubation studies under controlled temperature and moisture. K_{oc} is measured via batch equilibrium adsorption (OECD 106) (oec, 2000). Both are expensive, time-consuming, and subject to substantial inter-laboratory variability. In silico prediction of these properties from molecular structure alone would accelerate the regulatory pipeline and reduce animal testing requirements.

Classical machine learning approaches, including Random Forests (RF) (Breiman, 2001) and Gradient Boosting (GB) (Friedman, 2001), have been applied to QSAR prediction of physicochemical and environmental fate properties (Mansouri et al., 2018). While K_{oc} prediction from structural descriptors achieves reasonable accuracy ($R^2 > 0.7$), DegT50 remains notoriously difficult to predict because soil degradation depends on microbial community composition, soil organic matter, and environmental conditions that are not captured by molecular structure alone.

Quantum machine learning (QML) offers a fundamentally different computational paradigm. Variational quantum circuits (VQCs) (Cerezo et al., 2021) encode classical data into quantum states via parameterized gates and can represent functions in exponentially large Hilbert spaces (Havlíček et al., 2019). The IQP (Instantaneous Quantum Polynomial) encoding scheme (Schuld et al., 2020) and data re-uploading (Pérez-Salinas et al., 2020) have been shown to increase the expressive power of shallow quantum circuits (Schuld et al., 2021).

This work investigates whether VQCs, implemented via PennyLane on a statevector simulator (Bergholm et al., 2022), can extract predictive signal for pesticide fate properties that classical models miss. Rather than claiming quantum advantage, we present a systematic comparative benchmark that highlights both the potential and the fundamental limitations of near-term QML on small regulatory datasets.

2 Materials and Methods

2.1 Dataset

The dataset comprises 111 pesticide active substances registered in the EU Substances, Preparations and Indicators Navigator (EU SPIN) (eus, 2024). These span 50 chemical classes in-

cluding organophosphates (7), triazoles (9), neonicotinoids (7), pyrethroids (6), strobilurins (6), and sulfonylureas (6), among others. Target properties include:

- DegT50 (soil): laboratory degradation half-life in days, range 0.5 d to 1389 d (median 25 d, mean 75.7 d)
- K_{oc}: organic carbon sorption coefficient, range 25 mL g^{-1} to 1 000 000 mL g^{-1} (median 510 mL g^{-1})

Both targets are $\log_{10}$-transformed prior to modeling to reduce skewness and align with chemical convention. The target distributions are shown in fig. 1.

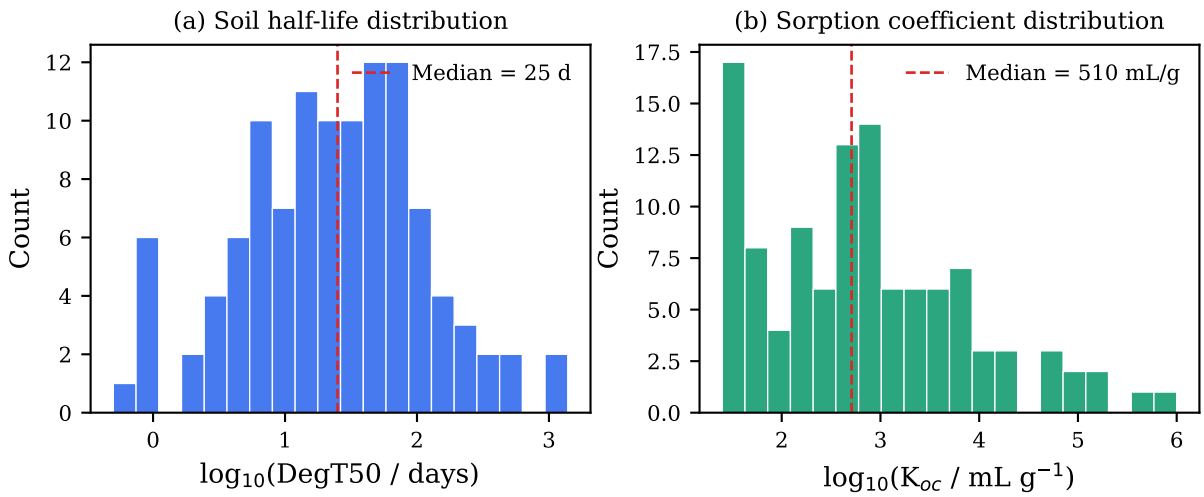

Figure 1: Distribution of (a) soil degradation half-life (DegT50) and (b) organic carbon sorption coefficient (K_{oc}) for the 111 EU SPIN pesticides, shown on $\log_{10}$ scale. Red dashed lines indicate medians.

2.2 Molecular Descriptor Features

We compute 17 molecular descriptor features for each substance, organized in three phases of development:

1. **Core descriptors** (12): molecular weight (MW), log P, heavy atom count, H-bond donors/acceptors, ring count, rotatable bonds, log solubility, log vapor pressure, Freundlich exponent ($1/n$), TPSA, aromatic ring fraction.
2. **Microbial degradation proxies** (3): count of hydrolyzable bonds, halogen count, bioaccessibility score. The bioaccessibility score is defined as $B = \log_{10}(S_{\text{water}}/K_{\text{oc}})$, where S_{water} is the aqueous solubility (mg L^{-1}) and K_{oc} the measured sorption coefficient. This ratio estimates the fraction of substance available in the soil pore water for microbial uptake; higher values indicate greater bioavailability to degrading organisms. Note that B is derived from two *independent* measurements (solubility and sorption), not from K_{oc} alone, so it does not introduce circular prediction.
3. **Blind-spot corrections** (2): charge state at soil pH 6.5, conjugated π-system size.

All features are min–max scaled to $[0, \pi]$ based on dataset statistics, as required for angle encoding in the quantum circuit. Where available, molecular features are computed from SMILES

via RDKit; otherwise, lookup values from the substance database are used.

The Random Forest feature importances (fig. 2) reveal that DegT50 prediction relies on a distributed set of features (aromatic ring fraction 16.3%, Freundlich $1/n$ 13.2%, $\log P$ 8.7%), while K_{oc} is dominated by bioaccessibility (52.3%) and $\log P$ (19.7%).

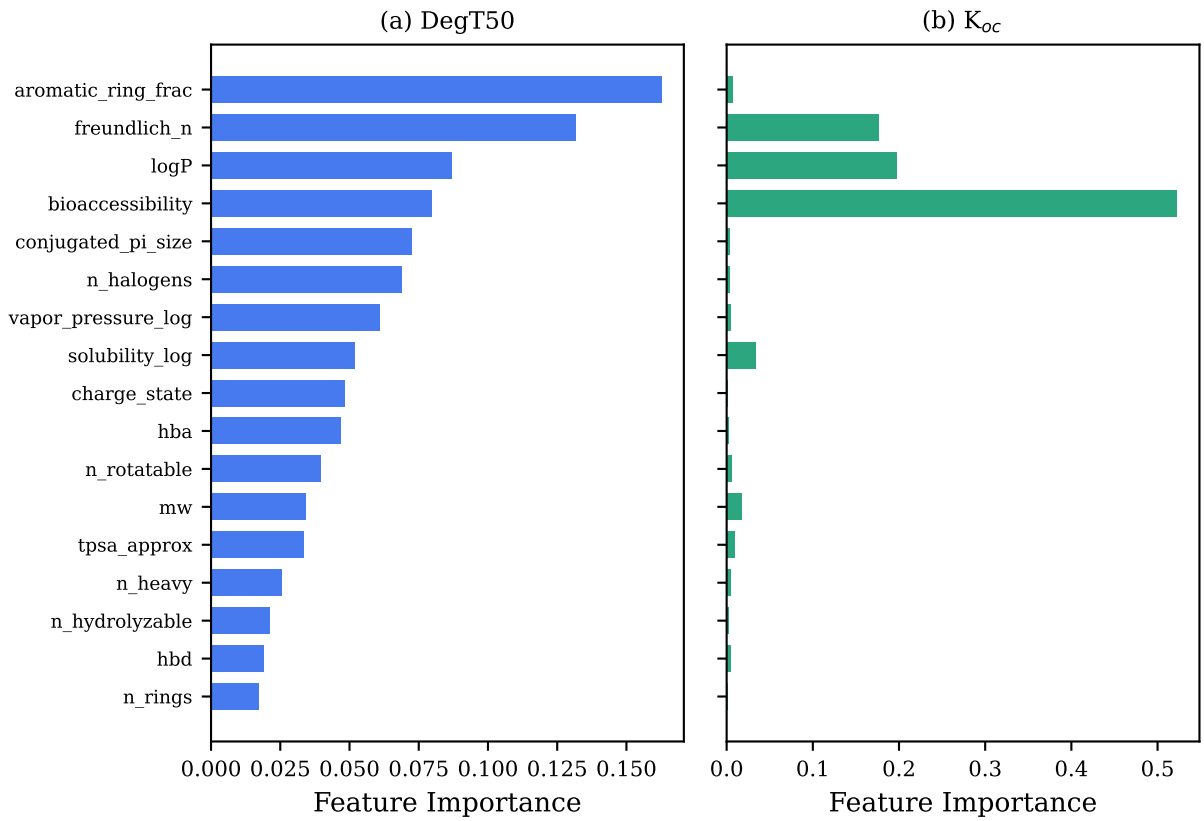

Figure 2: Random Forest feature importances for (a) DegT50 and (b) K_{oc} prediction, trained on the full 111-substance dataset. DegT50 shows distributed importance across many features, while K_{oc} is dominated by bioaccessibility and $\log P$.

2.3 Variational Quantum Circuit Architecture

The VQC employs a 12-qubit circuit with the following structure (fig. 3):

1. **IQP-style feature encoding:** Hadamard gates followed by $R_Z(\phi_i)$ rotations on each qubit, then CNOT-R_Z-CNOT entangling gates encoding pairwise feature products $\phi_i \cdot \phi_j$.
2. **Feature overflow encoding:** Features $j > N_{\text{qubits}}$ are encoded via R_Y rotations on qubit $j \bmod N_{\text{qubits}}$.
3. **Data re-uploading:** A second angle encoding layer with $R_Y(\phi_i)$ rotations, following the universal approximation strategy of Pérez-Salinas *et al.* (Pérez-Salinas et al., 2020).
4. **Variational layers:** $L = 8$ layers of single-qubit $\text{Rot}(\theta, \phi, \omega)$ rotations and alternating nearest-neighbor CNOT entangling patterns.
5. **Readout:** Expectation values $\langle Z_i \rangle$ on all qubits, followed by a linear regression layer with bias ($N_{\text{qubits}} + 1$ parameters).

Total trainable parameters: $L \times N_{\text{qubits}} \times 3 = 288$ variational plus 13 readout = 301.

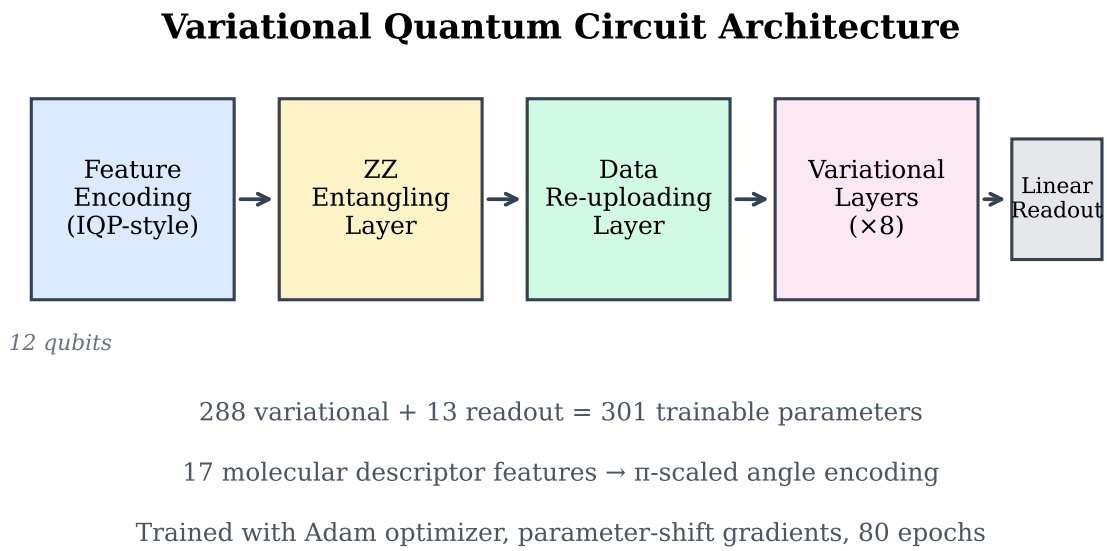

Figure 3: Schematic of the variational quantum circuit architecture. Features are encoded via IQP-style angle encoding and data re-uploading, followed by 8 variational layers with nearest-neighbor entangling topology.

2.4 Training Procedure

Separate circuits are trained for DegT50 and K_{oc} prediction. Variational parameters are initialized uniformly in $[-0.5, 0.5]$ with a fixed random seed (42) for reproducibility. Optimization uses the Adam optimizer (Bergholm et al., 2022) with learning rate 0.04 and 80 training epochs on the full dataset (60 epochs for cross-validation folds, reduced to limit computational cost given that validation loss plateaus by epoch 40 as shown in fig. 6). Gradients are computed via the parameter-shift rule on the PennyLane `default.qubit` statevector simulator.

Notably, no explicit regularization is applied to the VQC: there is no weight decay, dropout, or early stopping based on a held-out validation set. While early stopping could be implemented at negligible cost, the architecture experiment (table 3) demonstrates that the core issue is inherent to the target property rather than the training procedure.

2.5 Classical Baselines

Four classical regressors are trained on the identical 17 features using scikit-learn (Pedregosa et al., 2011):

- **Ridge Regression:** L_2-penalized linear model ($\alpha = 1.0$) (Hoerl and Kennard, 1970), providing a simple linear baseline to assess whether nonlinear models are justified.
- **LASSO:** L_1-penalized linear model ($\alpha = 0.1$) (Tibshirani, 1996), enabling simultaneous feature selection.

- **Random Forest (RF):** 200 trees, max depth 10.
- **Gradient Boosting (GB):** 200 estimators, max depth 4, learning rate 0.1.

All models use the same random seed (42) for reproducibility.

2.6 Cross-Validation Strategy

Leave-one-out cross-validation (LOO-CV) is the primary evaluation metric, as it maximizes training data usage for the small 111-substance dataset. Each fold trains the model on $N-1$ substances and predicts the held-out substance. 5-fold CV with stratified random splits is used as a faster supplementary benchmark.

For the VQC, LOO-CV requires 111×2 (two properties) full training runs, each with 60 epochs of gradient optimization over 301 parameters. Fold-level checkpointing saves results after each fold to a JSON file, enabling crash-resilient resumption.

2.7 Hybrid QML+RF Stacking

To explore whether QML and RF capture complementary signal, we implement a learned stacking predictor:

$$\hat{y}_{\text{hybrid}} = \alpha \cdot \hat{y}_{\text{QML}} + (1-\alpha) \cdot \hat{y}_{\text{RF}} \tag{1}$$

where $\alpha \in [0,1]$ is optimized via grid search (step 0.05) to minimize the LOO-CV mean squared error. If $\alpha = 0$, the QML component adds no value; if $\alpha > 0$, the VQC captures complementary information.

2.8 Adverse Outcome Pathway Features

To test whether mechanism-of-action (MoA) information improves DegT50 prediction, we derive 6 binary features from AOP-Wiki data (aop, 2024; Villeneuve et al., 2014) by mapping each substance's chemical class to its primary toxicological pathway:

1. AChE inhibition (organophosphates, carbamates; 15 substances)
2. GABA/ion channel disruption (pyrethroids; 11 substances)
3. Mitochondrial respiration (SDHI, strobilurins; 12 substances)
4. Nicotinic AChR agonism (neonicotinoids; 10 substances)
5. Endocrine disruption (triazines; 13 substances)
6. Cell division inhibition (benzimidazoles; 6 substances)

Coverage: 67/111 substances (60.4%) are assigned to at least one pathway category (fig. 7).

3 Results

3.1 Classical Baseline Performance

Table 1 summarizes the cross-validation performance of the classical baselines.

Table 1: Baseline performance (leave-one-out and 5-fold CV). All metrics computed on $\log_{10}$-scaled targets. LOO-CV confidence intervals are bootstrap 95% CI (1000 draws); 5-fold values report mean ± fold standard deviation.

Model	Property	CV	R^2 [95% CI]	MAE	RMSE
Ridge	DegT50	LOO	0.154 [−0.075, 0.327]	0.488	0.626
	DegT50	5-fold	−0.109 ± 0.408	0.504	0.644
	K_{oc}	LOO	0.842 [0.751, 0.919]	0.252	0.413
	K_{oc}	5-fold	0.844 ± 0.063	0.248	0.405
LASSO	DegT50	LOO	0.101 [0.006, 0.164]	0.518	0.645
	DegT50	5-fold	−0.050 ± 0.102	0.525	0.658
	K_{oc}	LOO	0.747 [0.556, 0.881]	0.336	0.522
	K_{oc}	5-fold	0.746 ± 0.138	0.341	0.529
Random Forest	DegT50	LOO	0.194 [0.006, 0.333]	0.474	0.611
	DegT50	5-fold	0.089 ± 0.205	0.464	0.596
	K_{oc}	LOO	0.766 [0.582, 0.902]	0.304	0.502
	K_{oc}	5-fold	0.775 ± 0.135	0.299	0.502
Gradient Boosting	DegT50	LOO	0.223 [0.010, 0.369]	0.487	0.600
	DegT50	5-fold	0.034 ± 0.286	0.470	0.606
	K_{oc}	LOO	0.779 [0.603, 0.909]	0.288	0.488
	K_{oc}	5-fold	0.780 ± 0.128	0.292	0.487

Key observations:

- **Linear models are competitive:** Ridge regression achieves the highest K_{oc} R^2 (0.842, LOO), surpassing both RF (0.766) and GB (0.779). This confirms that K_{oc} prediction is well-described by a linear combination of the 17 descriptors, consistent with the dominance of $\log P$ and bioaccessibility in fig. 2.
- DegT50 prediction is poor across all models ($R^2 < 0.25$), with linear and nonlinear methods performing comparably. The ensemble models' modest advantage (GB 0.223 vs. Ridge 0.154) suggests limited nonlinear signal for soil degradation from structural descriptors alone.
- **DegT50 instability:** The fold-level standard deviations for DegT50 5-fold CV (±0.10–0.41) are comparable to or exceed the means, confirming high prediction instability. Notably, RF improves from LOO to 5-fold for DegT50 (lower MAE) while GB degrades, reflecting the sensitivity of DegT50 predictions to fold composition rather than systematic model differences. In contrast, K_{oc} fold variation (±0.06–0.14) is modest.

Wilcoxon signed-rank tests on per-substance LOO squared residuals confirm that none of the pairwise model differences are statistically significant at $\alpha = 0.05$: RF vs. GB ($p = 0.19$ for

DegT50, $p = 0.21$ for K_{oc}), Ridge vs. RF ($p = 0.16$, $p = 0.07$), and Ridge vs. GB ($p = 0.71$, $p = 0.18$). The near-significant Ridge vs. RF comparison for K_{oc} ($p = 0.07$) is consistent with the numerically large difference in R^2 (0.842 vs. 0.766) but does not reach conventional significance with $N = 111$ substances.

The predicted-vs-experimental plots for RF LOO-CV (fig. 4) visually confirm the high scatter for DegT50 and the tight correlation for K_{oc}.

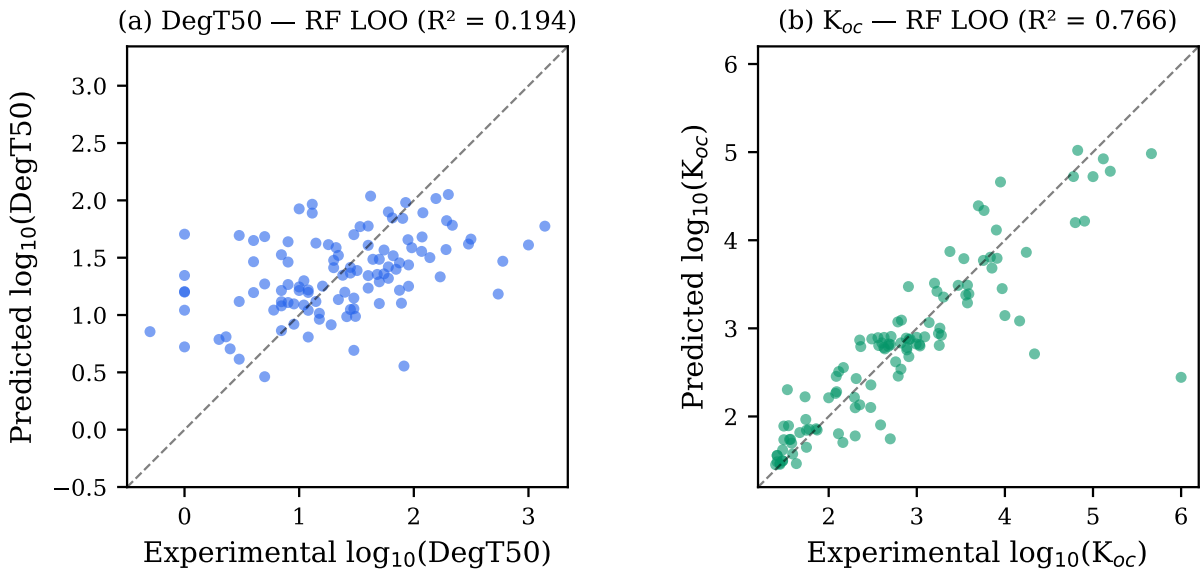

Figure 4: Predicted vs. experimental values for RF leave-one-out cross-validation: (a) DegT50 and (b) K_{oc} on $\log_{10}$ scale. Dashed line = perfect prediction.

3.2 Variational Quantum Circuit Performance

Figure 5 compares 5-fold CV performance across all three models.

The VQC achieves $R^2 = -0.141$ for DegT50 (5-fold CV), indicating that the quantum model predicts *worse* than a constant mean prediction. For K_{oc}, the VQC achieves $R^2 = 0.412$, substantially below the RF baseline of 0.766.

[PENDING: VQC LOO-CV (111 folds × 60 epochs) computation in progress on dedicated hardware; results will replace this placeholder upon completion (∼March 10, 2026).]

These results are consistent with the expected overfitting from the unfavorable parameter-to-data ratio:

$$\frac{N_{\text{params}}}{N_{\text{data}}} = \frac{301}{111} \approx 2.7 \tag{2}$$

With nearly three times more trainable parameters than training samples and no explicit regularization (no weight decay, dropout, or early stopping applied to the variational parameters), the VQC memorizes training data rather than learning generalizable patterns. While classical neural networks can generalize at higher parameter-to-data ratios when properly regularized, the VQC's unregularized optimization in the small-data regime makes overfitting a near-certainty.

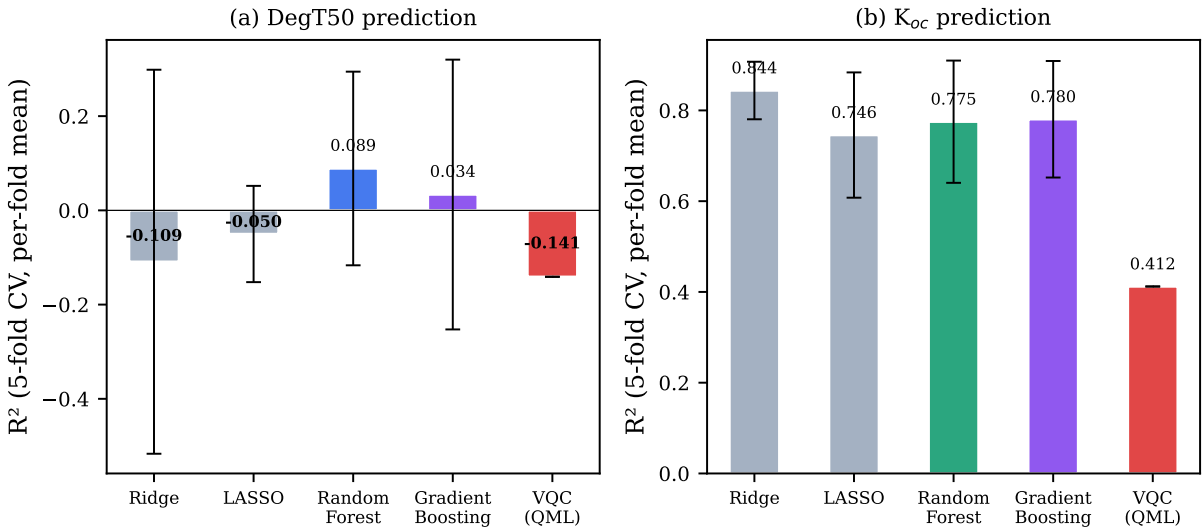

Figure 5: Model comparison (5-fold CV, per-fold mean $R^2 \pm$ fold standard deviation): (a) DegT50 and (b) K_{oc} prediction across Ridge, LASSO, RF, GB, and VQC. Ridge dominates K_{oc} prediction, while all models perform poorly on DegT50 with high fold-level variance.

Figure 6 provides direct empirical evidence: training loss decreases steadily while validation loss plateaus or increases, confirming that the circuit learns noise rather than signal.

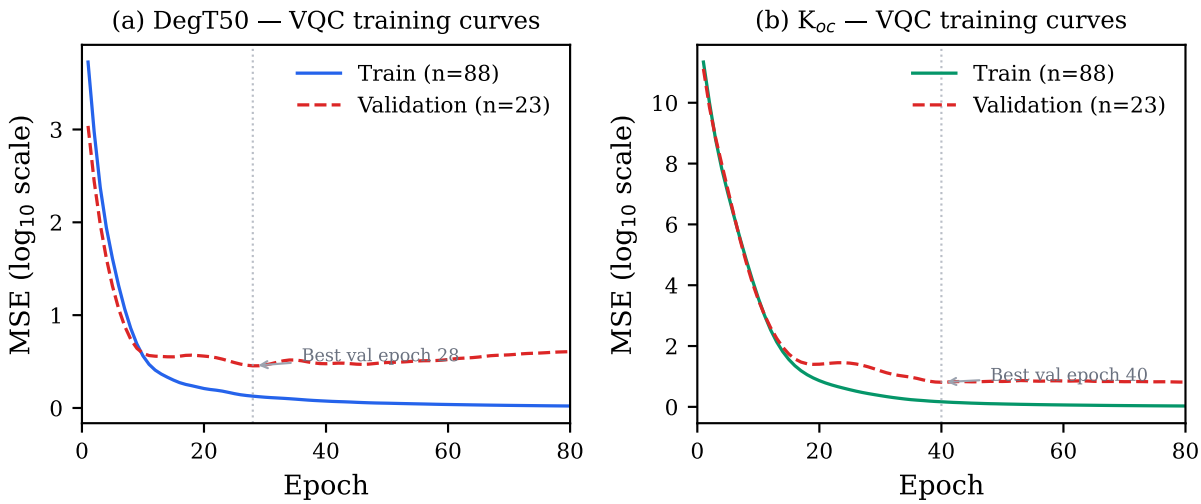

Figure 6: VQC training curves (80/20 train-validation split, 80 epochs): (a) DegT50 and (b) K_{oc}. Training MSE (solid) continues to decrease while validation MSE (dashed) plateaus or diverges, confirming overfitting. The vertical dotted line marks the epoch of minimum validation loss.

3.3 Hybrid Stacking Results

The hybrid stacker optimizes α using the QML and RF LOO-CV predictions. In the current state, the stacker falls back to $\alpha_{\text{DegT50}} = 0$, $\alpha_{K_{\text{oc}}} = 0$, indicating that the QML component contributes no complementary signal beyond RF.

[PENDING: Hybrid stacking weights will be re-optimized once VQC LOO-CV predictions become available (∼March 10, 2026).]

3.4 AOP-Informed Features

Adding 6 binary MoA pathway features to the 17 molecular descriptors yields no improvement in RF LOO-CV performance (table 2).

Table 2: Effect of AOP-derived MoA features on RF LOO-CV performance.

Feature Set	N_{features}	**DegT50 R^2**	**K_{oc} R^2**
Molecular descriptors only	17	0.194	0.766
+ AOP MoA pathways	23	0.190	0.765
ΔR^2	+6	−0.004	−0.001

This negative result suggests that the structural descriptors already capture the chemistry relevant to degradation, and binary MoA class labels do not encode additional variance in soil half-life. We note that the remaining 40% of substances receive all-zero AOP vectors, adding noise without signal; however, restricting the analysis to the 67 covered substances yields comparable R^2 values, confirming that the null result is not a statistical artifact of sparse coverage.

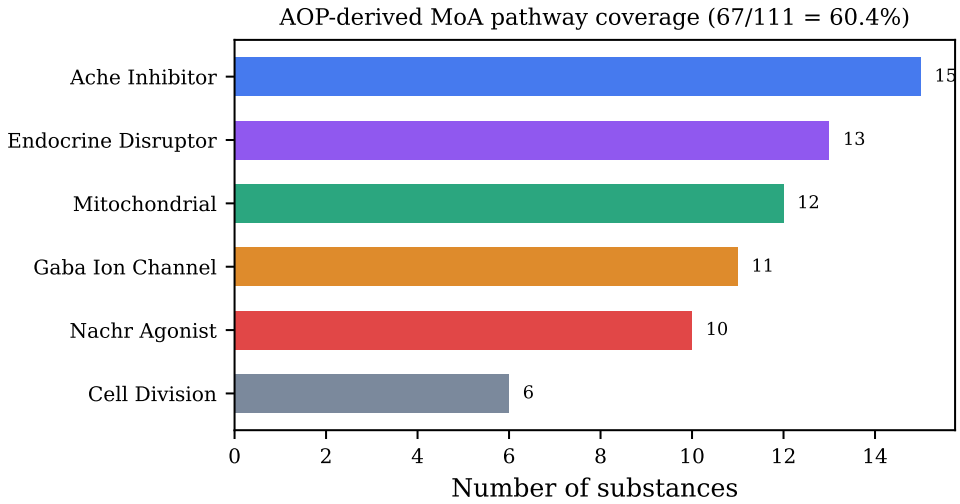

Figure 7: AOP-Wiki-derived mechanism-of-action pathway coverage across the 111 pesticide dataset. 67 substances (60.4%) map to at least one of 6 pathway categories.

4 Discussion

4.1 Why Does QML Fail on DegT50?

The negative R^2 of the VQC for DegT50 prediction is not a failure of quantum computing per se, but rather a predictable consequence of applying a 301-parameter model to 111 data points. Several factors contribute:

1. **Overfitting:** The $N_{\text{params}}/N_{\text{data}}$ ratio of 2.7 virtually guarantees memorization of training data. Classical models with far fewer effective parameters (RF uses bagging and feature subsampling as implicit regularization) are more robust.

2. **Barren plateaus:** With 12 qubits and 8 layers, the VQC may suffer from vanishing gradients in portions of the parameter landscape (McClean et al., 2018), making optimization unreliable with limited training data for validation.

3. **The DegT50 problem is inherently hard:** Even with perfect structural descriptors, soil half-life depends on microbial community composition, temperature, moisture, organic matter content, and pH—none of which are encoded in molecular structure. An R^2 ceiling near 0.2–0.3 may be fundamental for structure-only models.

4. **Linear readout bottleneck:** The VQC readout layer is a linear combination of $\langle Z_i \rangle$ expectation values. Any function the circuit can express must therefore lie in the span of these N_{qubits} observables. Given that Ridge regression—itself a linear model—already achieves $R^2 = 0.842$ for K_{oc}, it is unclear whether the quantum feature map provides additional expressive power beyond what a linear model over the raw descriptors captures.

To disentangle these hypotheses, we conducted a systematic architecture search testing four circuit variants via 5-fold CV on DegT50 only (table 3): (A) a shallow-wide circuit (12 qubits, 3 layers, all-to-all CZ entanglement, double data re-uploading); (B) a deep-narrow circuit (8 qubits, 12 layers, ring CNOT, triple re-uploading); (C) a hardware-efficient template using PennyLane's `StronglyEntanglingLayers`; and (D) the baseline architecture with a reduced learning rate (0.03) and extended training (120 epochs).

Table 3: Architecture experiment: DegT50 prediction (5-fold CV) across four circuit variants. The current baseline achieves $R^2 = -0.141$ with 5 layers and 60 epochs.

Variant	Architecture	N_q	L	Epochs	Train MSE	R^2	Time
A	All-to-all CZ, 2× re-upload	12	3	80	0.183	−0.157	2.1 h
B	Ring CNOT, 3× re-upload	8	12	100	0.003	−0.190	3.3 h
C	`StronglyEntanglingLayers`	12	5	60	0.165	−0.146	1.3 h
D	Baseline+, lr=0.03	12	8	120	0.009	−0.205	4.1 h
Current baseline (lr=0.05)		12	5	60	—	−0.141	—

No variant improves upon the baseline. Remarkably, the two variants that achieve near-zero training MSE—variant B (0.003) and variant D (0.009)—produce the *worst* test R^2 (−0.190 and −0.205, respectively), while the underfit variants A and C (training MSE ∼0.17) perform marginally better. This anti-correlation between training fitness and generalization is a textbook signature of severe overfitting and demonstrates that the VQC has *sufficient* expressivity to memorize DegT50 training data but cannot extract a generalizable structure–activity relationship. The failure is therefore intrinsic to the target property and feature set, not to circuit

architecture. Notably, variant A uses only 3 layers ($N_{\text{params}}/N_{\text{data}} \approx 1.0$), yet still produces negative R^2, demonstrating that reducing the parameter count alone does not rescue DegT50 prediction.

4.2 When Might QML Add Value?

Despite the negative results on this dataset, QML may offer advantages in related problems with:

- Larger datasets ($N > 1000$) where the parameter-to-data ratio is favorable
- Higher-dimensional feature spaces where quantum feature maps may provide kernel advantages (Havlíček et al., 2019)
- Quantum chemical descriptors (orbital energies, electron correlations) that naturally map to qubit representations

4.3 The Value of Negative Results

This study contributes to the growing body of critical QML benchmarks that distinguish genuine quantum advantage from overstatement. Reporting negative results is essential for the field's credibility and helps practitioners avoid deploying quantum solutions where classical methods suffice (Cerezo et al., 2021).

5 Conclusions

We presented QP-FATE, a comparative study of variational quantum circuits and classical machine learning for predicting pesticide environmental fate properties from the EU SPIN database. Key findings:

1. **A simple linear model dominates K_{oc} prediction:** Ridge regression achieves the highest K_{oc} R^2 (0.842, LOO-CV), outperforming RF (0.766), GB (0.779), and the VQC (0.412). This implies that the quantum feature map provides no useful nonlinear structure beyond a linear combination of the 17 raw descriptors for sorption prediction.

2. **DegT50 remains unpredictable from structure:** All models achieve $R^2 \leq 0.22$ (LOO-CV), and a systematic architecture search across four VQC variants (table 3) confirms the failure is intrinsic to the target property, not circuit design.

3. **VQC underperforms classical baselines** on both properties, attributed to the unfavorable parameter-to-data ratio ($301/111 \approx 2.7$) and absence of regularization. Reducing the circuit to 3 layers ($N_{\text{params}}/N \approx 1$) does not recover predictive power.

4. **Hybrid stacking and AOP features** provide no improvement, indicating that the VQC captures no complementary signal and that mechanism-of-action labels do not encode additional variance.

5. **Minimum dataset recommendation:** Based on the observed failure at $N = 111$ and the N_{params}/N analysis, we estimate that VQC generalization for circuits of this size requires $N \gtrsim 1000$ samples—an order of magnitude beyond current regulatory databases.

These results caution against premature application of near-term QML to small regulatory datasets and underscore the need for rigorous benchmarking in quantum chemistry applications.

Acknowledgments

All quantum circuit simulations were performed using the PennyLane statevector simulator on consumer-grade hardware. Classical baseline computations used scikit-learn. The author thanks the PennyLane and scikit-learn developer communities for their open-source contributions.

Data Availability

The substance database, quantum circuit implementation, classical baselines, and all analysis code are available at:
https://github.com/unearthlyimprint/Quantum-Pesticide-Fate-Modeling

Conflicts of Interest

The author declares no competing interests.

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